

Farah Zamir

AI Engineer · LLM Evaluation & Governance · Conservation & Preservation

London, UK · dynamicsofanaisteroid@gmail.com · GitHub · Substack · LinkedIn

AI Engineer building production evaluation infrastructure and LLM-powered systems at Virgin Atlantic, with academic rigour; studying the L6 AI Engineering Apprenticeship with Cambridge Spark. Background in public sector, information science, and digital preservation at a national scale - Parliamentary Archives, Science Museum Group, public libraries.

Operates in high-profile frontier AI in order to be best placed to challenge high-risk AI usage. Presented governance risks to senior executives and invited by Virgin Atlantic's VP of Data & AI to join a confidential OpenAI alpha channel for agent skills development - liaised personally with OpenAI technical partner to prevent Skills overreach. Hands-on experience with AI production; designed and shipped an internal AI/BI analytics tool adopted company-wide as the organisational standard. Architecting a multi-use AI Evaluation Harness using MLflow and LLM assistance to measure AI product outputs across business for analysis against fast-moving aviation data - self-nominated responsibility for rigorous guard-railing, human oversight to prevent AI risk, and documentation created **during**, not after production, in order to ensure a solid decision-making and audit record.

Motivated by background in sensitive data that AI is a neutral tool that can be used powerfully; this exponential power opens up a host of challenges that **must** be addressed before scale.

CORE EXPERIENCE

AI Engineer | **Virgin Atlantic** | Oct 2025 – Present

L6 AI Engineering Apprenticeship · Cambridge Spark (completing Jan 2027)

AI Evaluation Harness [Current — Major Project]

- Architecting a governance-first, standalone evaluation harness for LLM agent systems, intentionally decoupled from live production to establish a defensible trust layer before automation. Introduced human-in-the-loop as a **necessity** to this project rather than defaulting to Agent Skills (see Skills Contract below).
- Delivered as a seven-feature product pathway (F01–F07): data contract foundation for reliability and soundness, flowing downstream into intent evaluation engine, scoring framework, MLflow-instrumented execution, LLM query layer, stakeholder reporting, and handover/operating documentation.
- Feature 01 delivered: privacy-aware Spark pipeline over 7M+ observability spans, producing a canonical, evaluation-ready schema with sessionisation, DQ gates, redaction logic, and safe-field projection.
- Resolved a major telemetry parsing defect (NULL rate ~80% to ~20%) through source-structure validation; introduced cache/materialization strategy reducing downstream validation from ~23 minutes to sub-minute, consistent eye on compute usage and preventing unseen burn/consumption.
- Established explicit evaluation integrity controls: proxy GT vs human GT separation, hard quality gates, and refusal to relabel machine-consensus outputs as human-adjudicated ground truth.
- Added version-aware governance layer: dual-cohort evaluation, taxonomy diffing, field adapters, and fully version-tagged outputs with MLflow lineage, enabling full oversight and accountability.
- Technical stack: Databricks (Spark/Delta), MLflow, Azure DevOps, LLM prompt orchestration, taxonomy/rubric versioning, failure taxonomy, RAG-eval groundwork, prompt-injection test planning, and governance-by-design documentation.

Skills Contract & Risk Evaluation Audit [Prevention of major AI risk in above Harness, cross-business]

- Whilst in early stages of Harness, was given Databricks ai-dev-kit - a widely-deployed toolkit of Agent Skills across enterprise. Mindful of the recent OpenClaw crisis, I analysed a single skill and found wide backend permissions. I then carried out a full-scale audit, which identified undocumented attack vectors across the kit.
- Findings: malicious actors could author agent skills that cause users to execute destructive SQL operations using their own valid permissions, with no auth bypasses and no traces in the audit log distinguishing skill-generated code from user-written code. No published mitigation existed.
- I built the Skills Contract; a pre-execution governance framework covering enforced SQL projections, PII classification gates, audit trail requirements, and structured sign-off artifacts logged in MLFlow.
- Contract was validated against four agent skills and blocked production use on five of six critical criteria, and enclosed in Risk Evals document.
- Escalated immediately to VP and Head of Platform, governance meetings initiated. Both docs on GitHub.

AI/BI Genie — End-to-End Product Delivery & Governance

- Designed from the ground up an SQL-driven AI/BI tool that provided an LLM layer for engagement off of a highly-regimented core data table that redacted financial data and had huge permission limitations to avoid spill. Led full product life cycle, from creation to roll-out.
- Translated stakeholder requirements into prompt logic, SQL-backed retrieval flows, and benchmark question sets; achieved 100% accuracy on benchmark evaluation **before** allowing product usage.
- Created rollout gates to monitor product usage on a granular scale; used share permissions to ensure external usage was governed, and triage of Genie Instructions, SQL & Benchmarks for full internal protections against hallucinations or overreach.
- Appointed internal AI/BI Genie SME - supporting senior data colleagues across the organisation in troubleshooting and optimising deployments, including advising **against** Genie usage, when appropriate.
- Created 'Make Your Own Genie Kit' and Feedback & Version Control Logs to ensure consistency/rigor across build: Databricks notebooks, markdown documentation, slide deck templates, and feedback cycle frameworks. These were adopted as the organisational standard and integrated into the DevOps Wiki.

AI Chatbot Trainer | **Data Annotation AI** | Mar 2025 – Oct 2025

- Evaluated GPT and Claude model outputs for tone, accuracy, and safety alignment across real-world prompts
- Applied tagging and classification methodologies to LLM training datasets; developed working understanding of RLHF-adjacent evaluation workflows
- Built and tested custom GPT tools; gained foundational exposure to prompt engineering, model behaviour patterns, and most importantly, proprietary safety & ethical frameworks (or lack of them)

Database Assistant | **Sue Ryder** | Oct 2024 – Mar 2025

- Managed sensitive financial and charitable donor data in Raiser's Edge CRM; batch processing and rigorous error handling via Excel and FastStats.
- Built SharePoint technical process documentation; provided cross-team data support via Zendesk

Library Assistant | **Watford Central Library** | Sept 2019 – Aug 2024

- Maintained metadata and resource records using SPYDUS database and Dewey Decimal classification for thousands of annual users. In-person engagement with technical boundaries facing the public.
- Developed strong information governance, user-facing communication, structured data management instincts, and grounded philosophy in historical preservation, protection & dissemination of record.

Collection Care & Records Management Assistant | **Parliamentary Archives** | Sept 2017 – Jun 2018

- Supported classification and lifecycle management of Parliamentary Records under national archival standards, including records under seal. Academic study undertaken of chemical/environmental/human risks posed to collections, including CPD conferences at Kew and Churchill College, Cambridge
- Contributed to digital transition, SharePoint rollout, and cross-House compliance with the ARDP: foundational experience in regulated data governance and potentially catastrophic risk of data & trust breaches.

RESEARCH INTERESTS: AI FOR PUBLIC GOOD

Synapse Warehouse — Neuro-Inspired, Energy-Proportional AI Infrastructure

A proposed architectural departure from monolithic, always-on AI infrastructure, inspired by the human brain's ability to perform complex reasoning on approximately 20 watts, and mitigation of massive, irreversible resource consumption risks posed by global AI usage. Core concepts:

- Cortex Scrim: ultra-low-power orchestration layer performing intent classification to activate only the compute path a task requires, reducing idle consumption by an estimated 90%.
- Spiking Agent Nodes: Pre-baked agents powered on by event-driven 'spikes', designed using Neuromorphic Agent Engineering. This is the smart design AI Governed Layer that defines individual agents, maps node networks, switches "on" inter-relationships, and enforces kill switches & gating to prevent malicious actor misuse or energy overconsumption.
- Watermarking & Auditability Trail: Pre-built agents allow every output to be cryptographically signed/watermarked by that Synapse Node, including IP-to-IP detailed metadata. Every token produced by a Node can be linked to its training data, guardrails, and energy consumption.

Long-term aim: partner with renewable energy providers and neuromorphic hardware researchers (ARM, Intel Loihi) to enrich knowledge base and implement this project with green energy compute solutions, such as Follow-the-Sun.

Sub-categories of this interest remain at conservation and preservation scale; **environmental conservation** (AI-assisted camera-trap network for species tracking); **agricultural access** (grant navigation tooling bypassing bureaucracy for UK farmers post-Brexit); **information equity** (accessible AI interfaces for the elderly and digitally excluded users in libraries - services they are entitled to).